

Designer ECY1_2 Migration Tool 1.0.0

User Guide

DCAS - Distech / Designer Utility
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Document Purpose: This guide explains how to operate the Designer ECY1_2 Migration Tool 1.0.0, including package loading, .dg5 inspection, data point discovery, BACnet object matching, and saving updated migration files.

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Overview

Designer ECY1_2 Migration Tool 1.0.0 is a PowerShell/WPF utility designed to help technicians and support teams inspect Designer project package ZIP files, locate embedded .dg5 files, preview raw JSON content, extract data point paths, and assist with Gen 1 to ECLYPSE facilities migration workflows.

The tool provides a dark-themed graphical interface with package browsing, .dg5 listing, project filtering, raw JSON preview, data point extraction, BACnet object discovery, automatic replacement matching, and save-as functionality for modified .dg5 files.

Intended Users

Advanced support specialists, field technicians, application engineers, and internal support teams who work with Designer project packages, ECY/ECLYPSE controllers, BACnet object paths, and Gen 1 migration activities.

Users should be comfortable working with ZIP packages, controller IP addresses, credentials, and exported Designer project content.

Key Capabilities

- Browse and load a Designer package ZIP file.
- Scan the package and list all .dg5 files found inside the archive.
- Filter listed .dg5 files by project name.
- Preview selected .dg5 files as raw JSON without manually extracting the full package.
- Extract /Data... point paths from .dg5 JSON content into an editable data point table.
- Test HTTPS REST connectivity to an ECY/ECLYPSE host using Basic authentication.
- Discover local BACnet objects from supported object collections on the target controller.
- Automatically match extracted point names against discovered BACnet object names using fuzzy matching.
- Apply valid replacement paths and save the modified .dg5 as a new file.
- Open the selected .dg5 extraction location, copy internal ZIP paths, and open Designer in a browser.

System Requirements

- Windows operating system with PowerShell 5.1 or later.
- .NET Framework 4.5 or later with WPF support.
- Network access to the target ECY/ECLYPSE controller when using Test Connection or Auto-Fill Replacements.
- Valid controller credentials for REST API access.
- A Designer package ZIP file containing .dg5 files.
- PowerShell execution policy must allow the script to run in the current environment.

Before You Begin

- Work from a backup copy of the Designer package ZIP whenever possible.
- Confirm the target controller IP address or hostname is reachable from the workstation.
- Ensure the account used has sufficient permissions to access the controller REST API.
- Review any generated replacement values before applying and saving changes.
- Do not overwrite production files unless the migration result has been reviewed and validated.

User Interface Reference

Control / Area	Purpose
PACKAGE ZIP	Displays the selected Designer package ZIP path.
Browse...	Opens a file picker for selecting a ZIP archive.
Extract & List .dg5	Scans the selected ZIP and lists all embedded .dg5 files.
HOST / IP	Controller hostname or IP address used for REST API calls.
USERNAME / PASSWORD	Credentials used for Basic authentication to the controller.
Test Connection	Tests HTTPS access to /api/rest/v2/services on the target host.
Timeout (sec)	REST request timeout used during BACnet object discovery.
Auto-Fill Replacements	Discovers BACnet objects and fills replacement paths based on fuzzy name matching.
Show Discovered Objects	Displays the last discovered BACnet object name/path list.
Filter by Project	Filters .dg5 entries by project folder extracted from the ZIP path.
Preview Panel	Displays selected .dg5 raw JSON or data point table.
Wrap ON/OFF	Toggles word wrapping in raw JSON preview mode.
Data Points / Raw JSON	Switches between parsed data point view and raw JSON view.
Apply Replacements & Save	Applies valid path replacements and saves the modified content to a new file.

Standard Workflow

Step	Action	Details
1	Launch the Tool	Right-click the .ps1 file and select Run with PowerShell, or launch from an STA PowerShell session using: powershell -STA -File .\Designer-PackageExplorer.ps1
2	Select Package ZIP	Click Browse..., select the Designer package ZIP file, and confirm that the selected path appears in the PACKAGE ZIP field.
3	Extract & List .dg5	Click Extract & List .dg5. The grid populates with .dg5 file name, internal ZIP path, and file size.
4	Filter by Project	Use the Filter by Project drop-down to narrow the list to a specific project folder discovered from the ZIP path.
5	Preview a .dg5	Double-click a row or right-click and select Preview File. The side preview panel opens with the raw JSON content.
6	View Data Points	Click Data Points to switch from raw

		JSON to the parsed data point table. The tool extracts JSON path values beginning with /Data.
7	Test Controller Connection	Enter Host/IP, Username, Password, and click Test Connection. Status and console logs indicate whether the REST endpoint is reachable.
8	Auto-Fill Replacements	With a .dg5 preview open and data points loaded, click Auto-Fill Replacements. The tool queries BACnet object collections and attempts to match point names.
9	Review Matches	Review the Replacement column. Edit values manually if required. Rows marked (no match) are ignored during replacement.
10	Apply & Save	Click Apply Replacements & Save. Choose a destination filename. The tool writes a new modified .dg5 file using UTF-8 encoding.

Data Point Replacement Workflow

- Open a .dg5 file in the preview panel and switch to Data Points view.
- Confirm that the listed Full Path values are the original /Data... references to be migrated.
- Enter the target controller Host/IP, Username, Password, and Timeout value.
- Click Auto-Fill Replacements to discover BACnet objects from the controller.
- Review the Replacement column. The tool fills likely matches using exact or Levenshtein-distance fuzzy matching with a threshold of 5.
- Manually correct any path that was not matched or appears incorrect.
- Click Apply Replacements & Save and save the updated .dg5 as a new file.

Context Menu Actions

- Open Location in Explorer: extracts the selected .dg5 into a temporary Designer_Extracted folder and opens Windows Explorer to that file.
- Copy Path to Clipboard: copies the internal ZIP path of the selected .dg5 file.
- Preview File: opens the selected .dg5 in the side preview panel.
- Open in Designer: opens <https://<host>/designer> in the default web browser using the value entered in HOST / IP.

Output Files and Logs

- Modified .dg5 files are saved to the user-selected destination using the Save File dialog.
- The tool does not modify the selected ZIP file directly.
- The PowerShell console provides timestamped INFO, WARN, ERROR, and DEBUG messages.
- During BACnet discovery, raw REST responses or error bodies may be written to temporary JSON files for troubleshooting.

Troubleshooting

Issue	Likely Cause	Recommended Action
No .dg5 files found	Selected ZIP does not contain .dg5 entries.	Verify the package export and select the correct Designer project ZIP.
ZIP file not found	The selected ZIP was moved, renamed, or deleted.	Browse and select the package ZIP again.
Connection fails	Incorrect IP/hostname, network issue, TLS/HTTPS issue, or controller unavailable.	Confirm network reachability, IP address, HTTPS access, and controller status.
HTTP 401	Controller was reached, but credentials were rejected.	Verify username, password, and account permissions.
Auto-Fill returns no objects	REST endpoint unavailable, authentication issue, unsupported path, or no BACnet objects returned.	Use Test Connection, check credentials, and review the PowerShell console log output.
Many rows show (no match)	Point names differ significantly from BACnet object names.	Manually edit replacements or verify naming conventions before migration.
Apply Replacements warns no valid replacements	Replacement column is blank, unchanged, or contains only (no match).	Enter valid replacement paths or rerun Auto-Fill Replacements.
Modified file does not behave as expected	Incorrect path replacement or incompatible target object path.	Compare the original and modified .dg5, validate object paths, and test in a controlled environment.

Best Practices

- Always preserve the original Designer package and original .dg5 files.
- Use the Show Discovered Objects window to verify object names and paths before applying replacements.
- Treat fuzzy matches as recommendations, not guaranteed migration results.
- Review all replacement rows before saving the modified file.
- Perform validation in a test or staging environment before applying migrated files to production.
- Keep console logs available when troubleshooting connectivity or object discovery issues.

Known Limitations

- The tool bypasses SSL certificate validation for controller REST API calls. Use only on trusted internal networks and approved service workflows.
- Object discovery is limited to configured local BACnet collections: analog-inputs, analog-values, binary-inputs, binary-values, analog-outputs, and binary-outputs.
- Fuzzy matching uses point name similarity and may not always select the correct target object.
- The tool saves modified .dg5 content as a separate file; it does not automatically repackage the original ZIP archive.
- Only path values matching the /Data... pattern are extracted into the data point table.
- Controller REST API behavior may vary by firmware, configuration, permissions, and installed services.

Security Notice

- The script includes a global SSL certificate bypass for REST API calls. This is useful for internal controller diagnostics where self-signed certificates are common, but it reduces certificate validation protection.
- Use the tool only in authorized environments and on trusted networks.
- Avoid storing or sharing screenshots/logs that expose controller addresses, usernames, passwords, object paths, or customer site details.
- Passwords are entered through a PasswordBox in the UI, but credentials are still used in memory to build Basic authentication headers.

Disclaimer

Designer ECY1_2 Migration Tool 1.0.0 is provided as an internal support and migration assistance utility. It is intended to help inspect Designer package content and assist with path replacement workflows, but it does not replace engineering review, commissioning validation, or site-specific migration testing. Users are responsible for validating all generated replacements and confirming correct operation before use in production environments.

Revision History

Version	Date	Author / Owner	Description
1.0.0	July 1, 2026	DCAS / Advanced Support	Initial user guide for Designer ECY1_2 Migration Tool 1.0.0.